

warning devices that tell the pilot to “pull up” when he is too close to the ground, and others that warn of an impending stall or a collision course with another aircraft. There is an increasing tendency to go one step further and make the avionics automatically override the pilot if he is committing an error—pull the aircraft up when it is on a collision course with the ground, for example, or block a movement of the controls that might make the aircraft stall.

Creating a catastrophe

When accidents do happen, it is more often through a combination of factors than a single error or fault. The worst aviation disaster of the 20th century occurred in 1977 when two Boeing 747s, one operated by KLM and the other by Pan Am, collided on the ground at Tenerife in the Canary Islands, killing 582 people. The accident happened in poor visibility. Like many airports, Tenerife had no ground radar, so aircraft on the runways could only be tracked by visual observation, but the two 747s were invisible from the control tower, as well as to one another. The American aircrew, who had difficulty understanding the Spanish controller's English, were trying to follow instructions for taxiing, searching for the right turn off the runway. The Dutch KLM pilot, in a hurry and confused, assumed he had clearance to take off when he did not, and accelerated down the runway into the taxiing Pan Am 747. Weather conditions, substandard air-traffic control equipment, poor communications, and pilot error had all conspired to create a catastrophe.

AIR TRAFFIC CONTROL



IN ITS BASICS, THE SYSTEM for organizing air traffic devised in the 1930s and 1940s remains valid in the jet age. Before take-off a pilot files a flight plan and is cleared for a certain route by air-traffic control (ATC), which lays down the time and altitude at which he must pass waymarks along an airway. Each airliner has a “squawk” transponder, which emits a signal that identifies it to ATC radar, allowing a controller to track the aircraft. Controllers ensure that aircraft are separated vertically and horizontally—typically at least ten minutes flying time apart when on the same level, or 1,000 ft (300 m) apart by altitude.

The ATC centers that control the airways each cover a wide area, and may not even be situated at an airport. When an airliner draws

MODERN ATC ROOM

Computers have helped air-traffic controllers cope with the density of traffic, although air safety still depends on controllers' intelligence and steadiness under pressure.

near to its destination, it is handed over to approach controllers at the airport. They either place it in a holding pattern or in the line of landing aircraft. Approach controllers are in the airport control tower, but normally behind tinted glass because they work solely with radar screens. Aerodrome controllers, at the top of the tower behind clear glass, depend on radar and visual observation to deal with aircraft during taxiing, take-off, and landing. They take over as the aircraft makes its final approach and talk it down.

